Power system model decoupling for real-time simulation

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Introduction

The accurate decoupling of power system models is important when considering concurrent or parallel simulation. Decoupling allows bigger systems to be computed in real-time.

In power transmission systems, the presence of 'long' transmission lines allows the use of Distributed Parameter Line (DPL) models such as Bergeron line with losses or frequency-dependent line model (Wideband line model) that include explicit delays in their formulation. These delays, expressed in the z-domain by 1/z, allow for equation decoupling without any approximations. This technique in used since the mid-90s by RTDS Technologies in their power system simulators.

In distribution systems however, the typical line lengths are much less and do not allow the use of DPL model directly, that is because the propagation time is less than the simulation time step.

So let's consider the speed of light c (300000 km/s), and the simulation time step, Ts, in seconds.

Then, the minimum length of a line, MinLL =c*Ts.

For aerial lines, the fastest mode is actually very close to the speed of light. So, with a time step of 25 microseconds, MinLL=7.5km. The propagation speed in cable is typically much less than in aerial lines. Considering a cable propagation speed of 200000 km/s, MinLL=5 km.

In conclusion, DPL cannot be used to decouple the power system equations when dealing with networks with line length less than MinLL and DPL approximation model must be used.

The stubline

A stubline, in the context of power system simulation, is a special DPL model (i.e. a Bergeron line model with losses) in which the internal equations are tuned to obtain exactly a one time step propagation delay. Internally, it really is the same as a DPL.

The DPL propagation delay is computed with the following equations:

$$v = \sqrt{L * C}$$

$$\tau = l/v$$

L is the line inductance per unit of length (Henry/km); *C* is the line capacitance per unit of length (Farad/km); *v*= speed of propagation (km/s); *l* is the line length in km; *τ* is the propagation delay in seconds. For multi-phase lines, modal transformations produce similar model propagation delays.

In the stubline model, we instead set τ=Ts (the simulation time), then compute in reverse the DPL equation. In practice, a stubline will replace some existing inductance, typically a pi-line or a transformer leakage inductance. <u>The approximation of the model will consequently be that the stubline will add some equivalent capacitance and some errors in the simulation. Proper usage of stubline will minimized this error.</u>

Typical stubline usages are shown is the IEEE 13 node standard model (MATE_IEEE13Node.slx), namely Compensated multi-phase stubline and transformer with stubline. The use of both models allowed the accurate decoupling of the IEEE 13 node model into 3 subsystems that can be computed concurrently on a proper CPU-based real-time target, such as SpeedGoat.

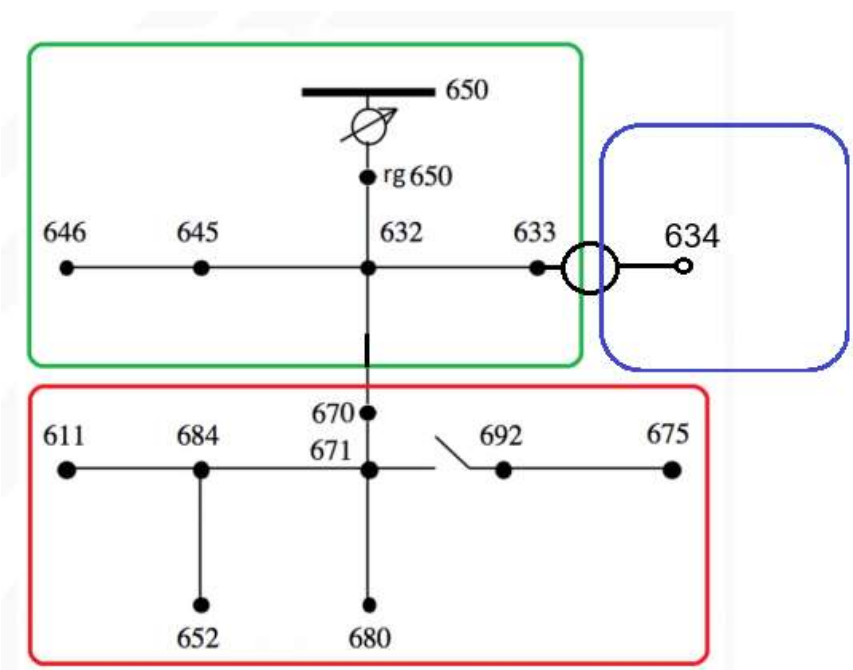

Figure 1 IEEE 13 node model

Stubline transformer

One of the most common usage of stubline is in combinaison with power transformers. By replace the secondary leakage inductance and resistance by a stubline having the same inductance value, we have created a clever way to decouple the power system at the transformer connection points.

This is shown is the IEEE 13 node standard model (MATE_IEEE13Node.slx), where the stubline transformer is between nodes 633 and 634.

In the SimScape Electrical model, the stubline transformer appears like in Figure 2, in two parts. This specific implementation allows Simulink Coder to actually generate separated tasks. It is also possible to to make the model appear in a single block, including then both sides of the stubline transformer. The single block option will not permit to build distinct executable tasks but still decouples the systems of equations in 2 distinct and smaller parts, thus accelerating the simulation.

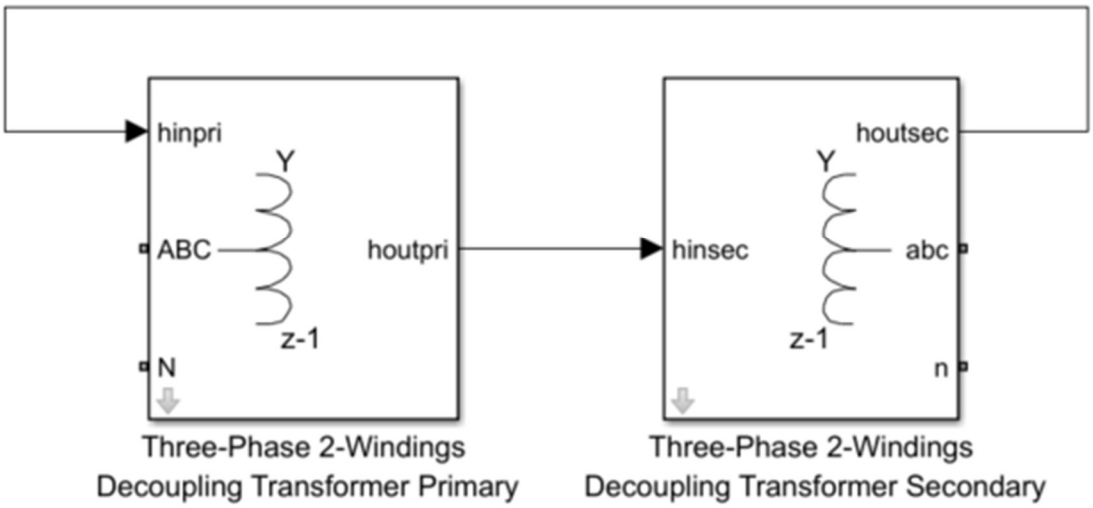

Figure 2: stubline transformer in 2 sections.

Compensated 3-phase stubline

In the same MATE_IEEE13Node.slx, a 3-phase Compensated Stubline Model (CTM) is used to decouple the grid between nodes 632 and 670-671, replacing the 2000 ft (0.61 km) line of the original model. The CTM has a one time step delay and the same total inductance; the internal stubline equation are then equivalent to a DPL of length of ~7km at 25 us. This is equivalent to adding spurious shunt capacitance (and reactive power) to the power system. The Compensated Stubline Model is designed to compensate for the reactive power at the power frequency by adding some shunt inductance. This will leads to accurate load flows as well as more accurate voltage and current amplitude and angle, something critical in PMU measurements for example. It is well known that in dense grids, voltage angles are very close to each other and small errors can lead to big problems.

Note that the 3-phase CTM is coupled, meaning that some phase-to-phase capacitance and mutual inductance in included in the resulting stubline. This is a better approximation than standard uncoupled 3-phase stubline which completely neglects inter-phase capacitances and inductance.

In addition to correcting the reactive power of the system, the CTM also produces more accurate transients responses during faults. To illustrate this, a single-phase-to-ground fault is made on bus 671 for test. Figure 3 compares the original pi-line to the standard and CTM cases. Shown is the main feeder input current (bus 632): Compensated Stubline Model response matches the case with a real 2000 ft pi-line. When using a standard non-compensated stubline, the response is erroneous.

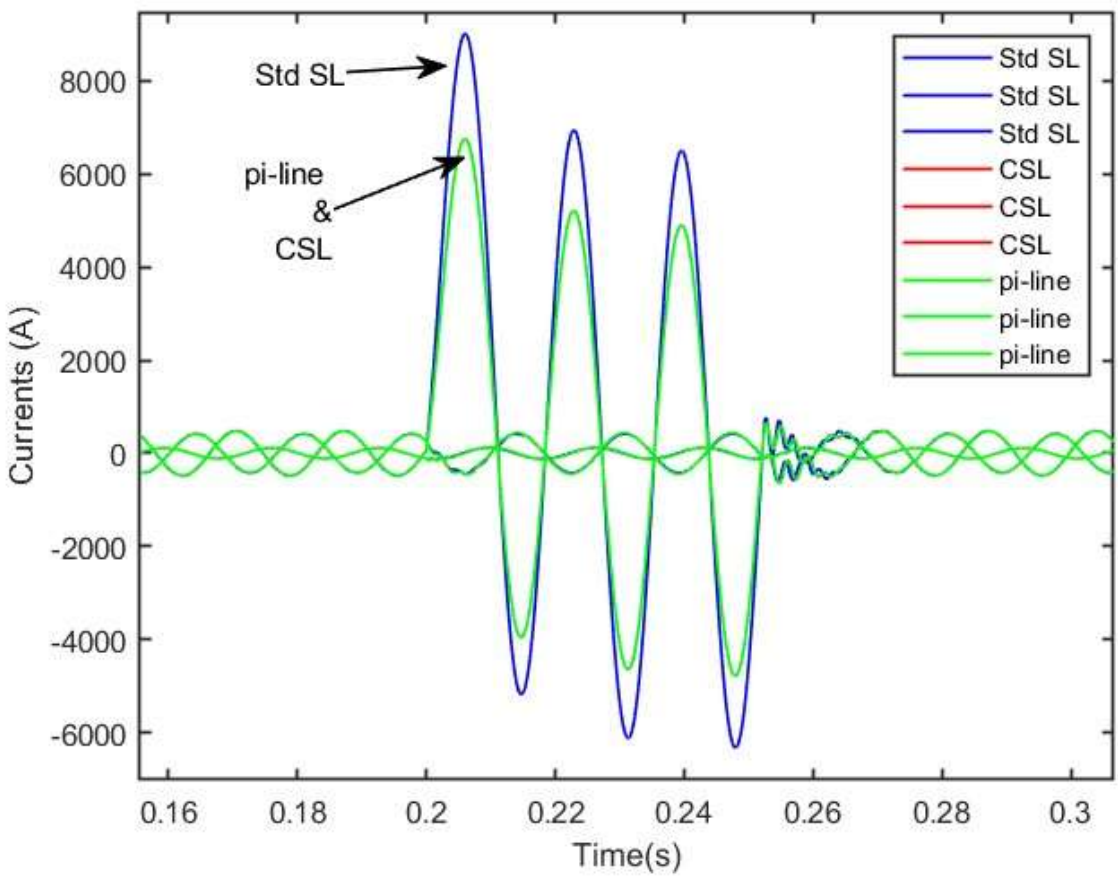

Figure 3 Main feeder input current for single-phase-to-ground fault is made on bus 671

References

[1] B. Ahmed, A. Abdelgadir, N. Saied and A. Karrar, "A Compensated Distributed-Parameter Line Decoupling Approach for Real Time Applications," in IEEE Transactions on Smart Grid, doi: 10.1109/TSG.2020.3033145.